INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Actapio, WA -- station KEAT, America/Los_Angeles
Building	OSM 813688319 OSM way 813688319
Facade gap	not applicable -- one building, no second facade
Weather record	43,269 real hourly records from KEAT
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-09-10 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. That is 6 more than the reactive incumbent, at 0 unsafe hours against its 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	1 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	21.670	21.0	21.670	0.847
01	mechanical	no	dry-bulb	21.390	21.0	21.110	0.769
02	mechanical	no	dry-bulb	21.390	21.0	21.110	0.720
03	FREE	yes	-	20.555	21.0	19.440	0.690
04	FREE	yes	-	20.555	21.0	20.000	0.658
05	FREE	yes	-	20.000	21.0	18.890	0.575

06	FREE	yes	-	18.890	21.0	17.780	0.575
07	FREE	yes	-	20.830	21.0	19.440	0.895
08	FREE	yes	-	20.555	21.0	17.780	1.300
09	FREE	yes	-	20.835	21.0	18.330	1.284
10	mechanical	no	dry-bulb	21.385	21.0	17.220	1.212
11	mechanical	no	dry-bulb	21.390	21.0	18.330	1.403
12	mechanical	yes	-	20.555	21.0	16.110	1.476
13	mechanical	yes	switch budget	19.445	21.0	15.560	1.450
14	mechanical	yes	switch budget	19.445	21.0	15.560	1.286
15	mechanical	yes	switch budget	17.780	21.0	16.110	0.943
16	mechanical	yes	switch budget	17.225	21.0	16.670	0.892
17	mechanical	yes	switch budget	16.390	21.0	16.110	0.879
18	mechanical	no	dew point	15.555	21.0	15.000	0.847
19	mechanical	no	dew point	16.115	21.0	15.560	1.272
20	mechanical	no	dew point	15.835	21.0	15.560	1.477
21	mechanical	no	dew point	15.280	21.0	15.560	1.392
22	mechanical	yes	-	15.560	21.0	15.560	1.180
23	mechanical	yes	-	15.835	21.0	16.110	0.959

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 21.0 C limit, so it fails by 0.670 C. It would take a limit 0.670 C higher, or a bound 0.670 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 21.0 C limit, so it fails by 0.390 C. It would take a limit 0.390 C higher, or a bound 0.390 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 21.0 C limit, so it fails by 0.390 C. It would take a limit 0.390 C higher, or a bound 0.390 C tighter, to change this hour.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.555 C, which is 0.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.689 C, and the margin is measured, not chosen: 0.689 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.555 C, which is 0.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.658 C, and the margin is measured, not chosen: 0.658 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.000 C, which is 1.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.575 C, and the margin is measured, not chosen: 0.575 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.890 C, which is 2.110 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.575 C, and the margin is measured, not chosen: 0.575 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.830 C, which is 0.170 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.895 C, and the margin is measured, not chosen: 0.895 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.555 C, which is 0.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.300 C, and the margin is measured, not chosen: 1.300 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.835 C, which is 0.165 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.284 C, and the margin is measured, not chosen: 1.284 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.385 C against a 21.0 C limit, so it fails by 0.385 C. It would take a limit 0.385 C higher, or a bound 0.385 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 21.0 C limit, so it fails by 0.390 C. It would take a limit 0.390 C higher, or a bound 0.390 C tighter, to change this hour.

12:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.555 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.445 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.445 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.780 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.225 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.390 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.555 C would have passed. The outside air is too HUMID: the dew-point bound is 15.23 C against a 15.0 C maximum, failing by 0.23 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.115 C would have passed. The outside air is too HUMID: the dew-point bound is 15.28 C against a 15.0 C maximum, failing by 0.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.835 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.280 C would have passed. The outside air is too HUMID: the dew-point bound is 15.28 C against a 15.0 C maximum, failing by 0.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.560 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.835 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.